20.3. BREEDER REACTORS

The possibility to breed excess fissile nuclei by consumption of fissile + fertile nuclei increases with increasing fission factor η according to §19.9. Since η depends on the fission cross section, which is highly energy dependent (Fig. 19.3), the neutron spectra in reactors must be considered. In a reactor with large amounts of highly moderating atoms, the original un-moderated spectrum is strongly shifted towards lower energies. However, even under optimal moderating conditions, a thermal Maxwell neutron distribution is never achieved in a reactor, but can be closely approached in thermal columns of research reactors. Because breeding occurs through neutron capture in fertile ^{232}Th and ^{238}U (the *only* fertile atoms in nature), with maximum probability between 600 and 700 eV, it is desirable for breeding that the neutron flux not be too low in this region. Just as breeding cannot be achieved with a thermal Maxwell neutron distribution it cannot be achieved with a fission neutron energy spectrum. Even if attempts are made to produce very "hard" neutron spectra (i.e. high energy), some moderation always occurs. Thus in a fast reactor, in which as few as possible light atoms are used, the neutron spectrum is still shifted strongly towards a lower energy region (Fig. 19.4) so that typically the neutron flux at 0.25 MeV is higher than that at 1 MeV. The curves shown for thermal and fast reactors in Figure 19.4 are only indicative, depend strongly on the particular reactor and fuel design, change with time, and vary over the core.

Because breeding is only possible for $\eta > 2$, ^{233}U is the favored fissile material for breeding in thermal reactors, and ^{239}Pu for breeding in fast reactors. Considering the shift towards lower neutron energies in fast reactors it is seen that η for ^{239}Pu is always greater than for ^{233}U. Thus ^{239}Pu can be more rapidly obtained by breeding than ^{233}U. Consequently much more effort has been put into the development of fast plutonium breeders than into thermal ^{233}U breeders.

20.3.1. The thermal breeder

In the thermal breeder reactor, ^{233}U is produced from ^{232}Th. The reactor may be designed either with a core containing a mixture of ^{232}Th and ^{233}U, or with a central zone (*core*) of ^{233}U surrounded by an outer layer (*blanket*) of ^{232}Th. However, it is necessary to minimize parasitic neutron capture in structural materials including monitoring systems, control rods, etc. Calculations have shown that a conversion ratio of 1.06 should be possible.

As an alternative to using a solid fuel matrix, both core and blanket can be liquid. In the HRE-2 (homogeneous reactor experiment), USA, the core consisted of a ^{233}U sulfate solution in D_2O, while the blanket was a suspension of ThO_2 in D_2O. The project was terminated because of various technical difficulties.

The molten salt reactor (MSR) was a somewhat similar design tested at Oak Ridge, USA, for some years. Here UF_4 and ThF_4 were dissolved in an eutectic melt of BeF_2, ZrF_4, and LiF (or NaF). The reactor may be homogeneous with respect to ^{233}U and Th, or of the core and blanket type in which case graphite may be used as moderator. The molten salt system has several advantages: (i) radiation effects are less in molten salts than in solid fuels, (ii) the expense of fabricating complicated fuel elements is avoided, (iii) the fission products can be removed continuously, (iv) fuel recharging can be made continuous, (v) the core contains little absorbing structural materials, and (vi) the system has a high negative temperature coefficient and is therefore easy to control. Points (iii) and (v) makes it possible to obtain a conversion ratio of 1.07, i.e. a 7% breeding gain; the fissile doubling time has been estimated to be 20 y. In the MSR high salt temperatures ($\leq$700°C) were used. Nevertheless, the system could be operated near atmospheric pressure because of the low vapour pressure of the melt. The salt is also inert to air and water. By using a dump tank for the core, the risk of criticality accidents was minimized. The on-line fission product removal required novel chemistry, and these problems were never fully solved.

Figure 20.4 shows the production of heavy atoms in the ^{232}Th system. Because ^{233}Th is produced until the fuel is removed from the core (or blanket), the long half-life of ^{233}Pa (27 d) requires long cooling times before reprocessing if a maximum yield of ^{233}U is to be obtained. Furthermore, when continuous on-line reprocessing with removal of ^{233}Pa is not used, the decay of ^{233}Pa will lead to a slow increase in reactivity with time after

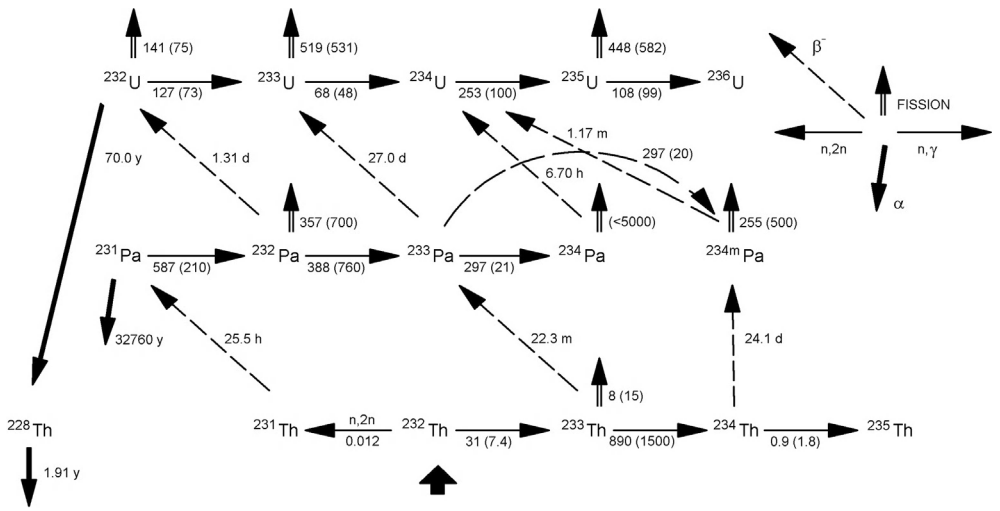


Figure 20.4 Nuclear reactions in irradiated thorium. Numbers are half-lives or effective reaction cross-sections for a power LWR; 0.025 eV data are given within parentheses.

shutdown due to a slightly increasing amount of ^{233}U from decay of ^{233}Pa. The thorium would become contaminated by the isotope ^{234}Th and 240 d would be required to reduce this activity to 0.1%. The ^{233}U would be contaminated by ^{234}U and ^{232}U; the latter constitutes a problem because of its relatively rapid decay into the $^{228}\text{Th} \rightarrow {}^{224}\text{Ra} -$ chain. These isotopic contaminations of the fertile and fissile products in the $^{232}\text{Th}-{}^{233}\text{U}$ fuel cycle would cause handling difficulties, which are more severe than for the $^{238}\text{U}-{}^{239}\text{Pu}$ fuel cycle. At present no ^{233}U breeder exists.

Several research groups have argued that a subcritical thorium breeder, driven by an accelerator based high intensity neutron source, might be a good and safe future energy producer. Using a molten salt/thorium fuel and coolant combination in continuous loops through a D_2O moderator tank or a graphite moderator it could generate a relatively high power from a small inventory of fissile material. It is also possible to design a fast, lead/bismuth cooled, Th-based subcritical breeder by using an intense external n-source. However, no such devices have yet been built and tested.

20.3.2. The fast breeder

The principal design of a fast breeder reactor consists of a central core of plutonium in which fission occurs, surrounded by an outer blanket of ^{238}U in which neutrons are captured to form new ^{239}Pu. This blanket is surrounded by a reflector, usually of iron. The fission yield curve of ^{239}Pu is similar to that of ^{235}U (see Fig. 14.9) but its heavy mass peak is shifted up by a few atomic mass units. Some neutron capture occurs in ^{239}Pu yielding fertile ^{240}Pu, which through another neutron capture produces fissile ^{241}Pu; similarly, successively higher elements are formed in time. In a strong neutron flux their fate is destruction by fission. Thus, in the long run, by conversion of ^{238}U into ^{239}Pu, etc., fission energy is always released and, in principle, all the ^{238}U can be used for producing fission energy. In practice a value of $\sim$70% is considered more realistic. Still this would mean $\sim$100 times more energy than is available from fission of ^{235}U only. Since the fast breeder reactor concept extends the fission energy resources by about a factor of 100, using a proven technology, it makes uranium the largest fossil energy resource presently available on earth. In addition fast reactors may also be used to fission long lived minor actinides such as americium and thereby not only increase the energy usage of the fuel but also shorten the longlevity of the final nuclear waste. Thus, if a separation process is designed to recover both uranium and plutonium together with the minor actinides a new fuel could be made directly using this composition. This is often called homogenous recycling.

The conversion efficiency of a reactor of this design can be estimated from

$$C = \eta + \varepsilon' - p' - \Lambda' - 1 \tag{20.2}$$

where ε' is the contribution to the neutron production from fertile nuclides, p' is the neutron loss due to parasitic absorption, and Λ' due to leakage. Some representative values are

	η	ε'	p'	Λ'	C
Core	2.56	0.27	0.049	1.15	0.63
Blanket	–	0.10	0.020	0.032	–
Total	2.56	0.37	0.069	0.032	1.83

It should be observed that the neutrons lost from the core are caught by the blanket. C is higher than in any other known chain-reacting system.

The longest experience with fast research reactors has been provided by the BN-350 reactor in the former USSR, now Kazakhstan. It was operated at a power level up to 1000 MW_{th} from 1972 to 1994. Most of the heat was used for water desalination (120 000 m^3/day) and only a small amount for electricity production. Larger (100–1200 MW_e) prototype fast breeder power reactors have been designed in the former USSR, France, the UK, and Japan. The largest prototype FBR built and operated so far is the Creys-Malville plant (located in France), which could produce 1 200 MW_e.

Figure 20.5 shows the Creys-Malville reactor; in operation from 1986 to 1997. The thermal rating was 3000 MW_{th} (gross 1240 MW_e); the neutron flux was very intense, $\leq 6 \times 10^{19}$ n $m^{-2}s^{-1}$, which puts a severe strain on construction materials. The core consisted of an inner and an outer fuel zone surrounded by a blanket, a steel reflector and a neutron shield, all immersed in liquid sodium (3 300 t) contained in a 25 cm thick austenitic steel vessel at atmospheric pressure. The reactor vessel was enclosed in a safety vessel and surrounded by a cooled concrete (biological) shield. The fuel was made of pins placed in hexagonal shrouds. The fuel zones contained 15 vol.% PuO_2 and 85 vol.% depleted UO_2, while the breeding blanket only contained depleted UO_2 (0.2% ^{235}U). The fuel pins were clad in 0.5 mm stainless steel. The total plutonium amount was 4 800 kg. The neutron shield consisted of hollow steel tubes. Because the sodium becomes extremely radioactive due to ^{24}Na formation, the whole primary cooling system was contained in the reactor vessel. By means of a sodium/sodium intermediate heat exchanger within the reactor vessel (but outside the neutron flux) heat from the primary circuit (542°C) was transferred to a secondary sodium stream (525°C) and transported to a sodium/water steam generator which produced 487°C steam at 17.7 MPa. Heat exchanger and piping were made of steel.

The prompt neutron lifetime in a fast reactor is about 1000 times smaller than in a thermal reactor. A reactor which can go critical on the prompt neutrons only would be exceedingly difficult to control. Therefore, fast reactors are designed to depend on the delayed neutrons (like thermal reactors). The time period (t_{per}) is large enough to allow reactor control through the use of neutron absorbing rods. Since the neutron spectrum is

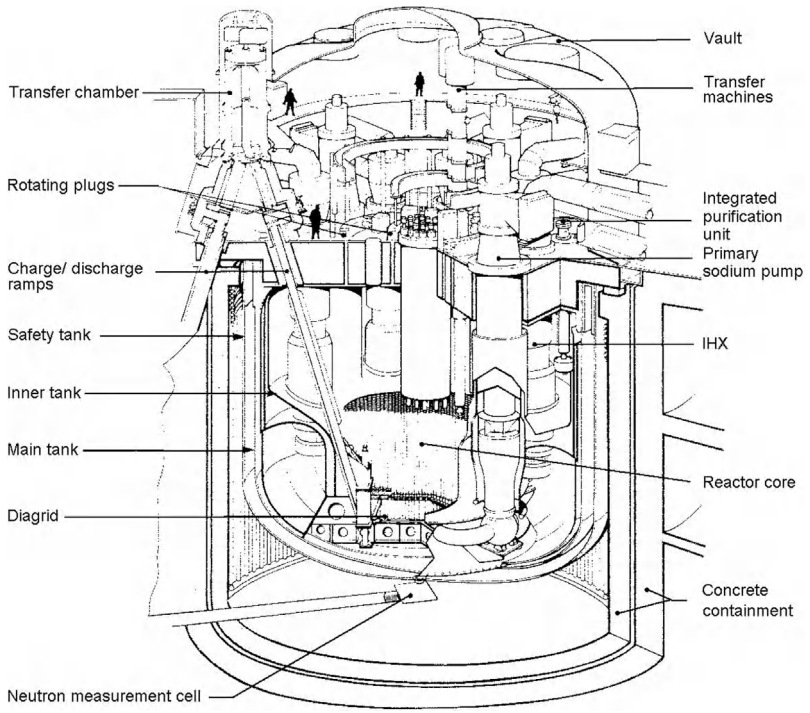


Figure 20.5 Vertical section of European prototype fast power reactor Creys-Malville (Superphenix). The vessel was filled with 3 300 t of liquid sodium.

such that several percent of the flux is in the resonance region, control rods with boron can be used; in practice boron carbide (the carbon atom reduces the neutron energy further) and/or tantalum (which has large absorption peaks $\geq 10^4$ b at 3–100 eV) are used.

Although the temperature coefficient of the Creys-Malville reactor was negative, this is not so for fast Pu-fuelled reactors with harder neutron spectra, e.g. the Russian BN-350. If the fast reactor becomes overheated, the core could be deformed, making it prompt critical. The power would then increase rapidly with a doubling time in the microsecond range, and a severe accident would be unavoidable. To prevent this the core is designed to achieve negative reactivity upon sudden power transients by using $^{238}UO_2$ in the core. When the temperature rises, doppler broadening occurs in the ^{238}U resonance capture region, and consequently more neutrons are consumed by ^{238}U, which limits the power increase. The shorter the neutron lifetime, the less would be the power excursion. In addition the thermal expansion of the fuel reduces the reactivity, which makes the temperature coefficient smaller or more negative.

The total breeding ratio in the Creys-Malville reactor was 1.24. In BN-350 a breeding ratio of 1.5 was obtained because it had less neutron absorbing materials. Still higher breeding gain can be achieved in fast reactors at some sacrifice of safety control.

Considerable radioactivity is induced in the sodium in the primary cooling circuit of a liquid metal cooled fast breeder reactor (LMFBR): $^{23}Na(n,\gamma)\,^{24}Na$, $^{23}Na(n,p)\,^{23}Ne$, etc. ^{24}Na has a 15 h $t_{½}$ and emits energetic γ's. The primary cooling loop must therefore be well shielded. Its activity is a nuisance only in case of repair work in the primary system, requiring considerable waiting time before the loop can be approached. The sodium dissolves many of the corrosion and fission products eventually released. To remove these, the primary loop is provided with cold trap purification systems.

Other coolants currently discussed, e.g. in Europe, is molten lead and helium. For all coolants there are advantages and disadvantages. For example lead is excellent for accident management since a core melt will dissolve in the lead and thus disarm the situation leaving a large, heavily contaminated lead block. Also some of the more troublesome volatile fission products such as iodine will react with the hot lead and be trapped preventing their release to the environment. A considerable disadvantage with using lead as a coolant is the increased corrosion in pumps and pipes. Sometimes bismuth is added to the lead in order to lower the melting point and make the environment less aggressive. This will, however, produce polonium during reactor operation. Any release of gas from a gas-cooled reactor will contain some radioactive iodine formed from tramp uranium and leaking fuel. Thus special venting systems and filters will be needed.

20.4. REACTOR WASTE

In the "ideal" nuclear reactor all fission products and actinides produced are contained in the fuel elements. As mentioned in §19.17, there are four processes in a water cooled reactor through which radioactivity leaves the reactor vessel; in all cases the carrier of activity is the coolant:

(i) induced radioactivity in the cooling medium (and moderator if separate);
(ii) corrosion products containing induced activities (mainly from materials in the feed water);
(iii) leaked fission products and actinides from faulty fuel elements;
(iv) fission products of actinides deposited on surfaces in the core ("tramp uranium").

We order the discussion of reactor wastes according to the phase which contains the radioactivity (gas, liquid or solid) and end by a brief discussion of waste from decommissioning of nuclear reactors.

20.4.1. Gaseous wastes

The intense flux of fast and thermal neutrons induces several radioactivities in H_2O: $^{2}H(n,\gamma)^{3}H$, $^{16}O(n,p)^{16}N$, $^{18}O(n,\gamma)^{19}O$, and $^{18}O(n,p)^{18}F$. Considerable activities of ^{13}N are produced through the reactions $^{13}C(n,p)^{13}N$ and $^{14}N(n,2n)^{13}N$ with carbon in the